4.5.2 Crystal Field Theory

Valance bond theory helps us to visualise the bonding in complexes. However, it has limitations as mentioned above.Hence Crystal Field Theory to expalin some of the properties like colour, magnetic behavior etc.,This theory was originally used to explain the nature of bonding in ionic crystals. Later on, it is used to explain the properties of transition metals and their complexes. The salient features of this theory are as follows.

1. Crystal Field Theory (CFT) assumes that the bond between the ligand and the central metal atom is purely ionic. i.e. the bond is formed due to the electrostatic attraction between the electron rich ligand and the electron deficient metal.

2. In the coordination compounds, the central metal atom/ion and the ligands are considered as point charges (in case of charged metal ions or ligands) or electric dipoles (in case of neutral metal atoms or ligands).

3. According to crystal field theory, the complex formation is considered as the following series of hypothetical steps.

Step 1: In an isolated gaseous state, all the five d orbitals of the central metal ion are degenerate. Initially, the ligands form a spherical field of negative charge around the metal. In this filed, the energies of all the five d orbitals will increase due to the repulsion between the electrons of the metal and the ligand.

Step 2: The ligands are approaching the metal atom in actual bond directions. To illustrate this let us consider an octahedral field, in which the central metal ion is located at the origin and the six ligands are coming from the +x, -x, +y, -y, +z and -z directions as shown below.

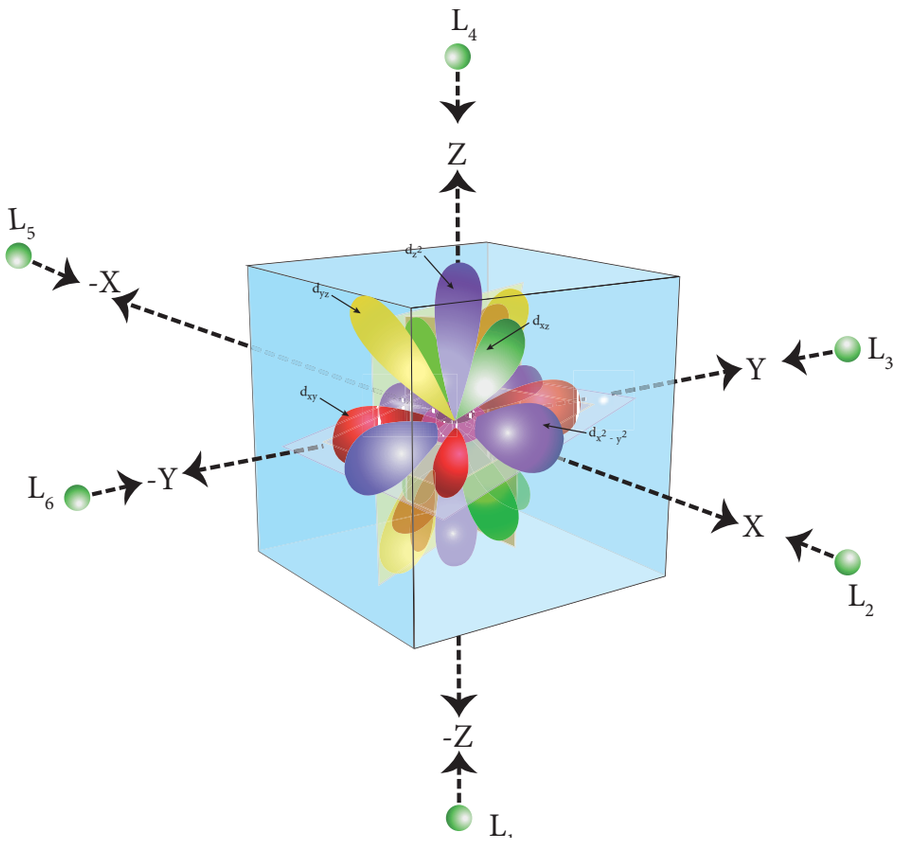


Figure 5.10 octahedral ligand field